

Version Control, IDEs, and Documentation



Using Git and Markdown in VSCode

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Disclaimer

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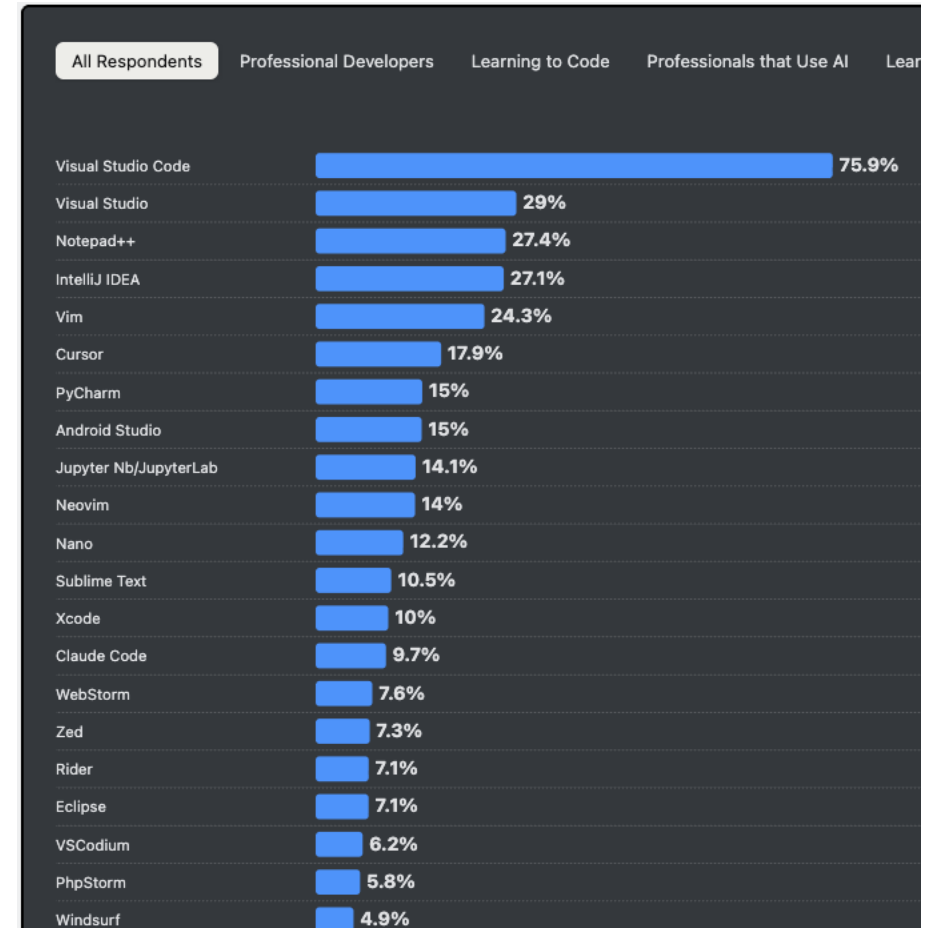
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Module Objectives

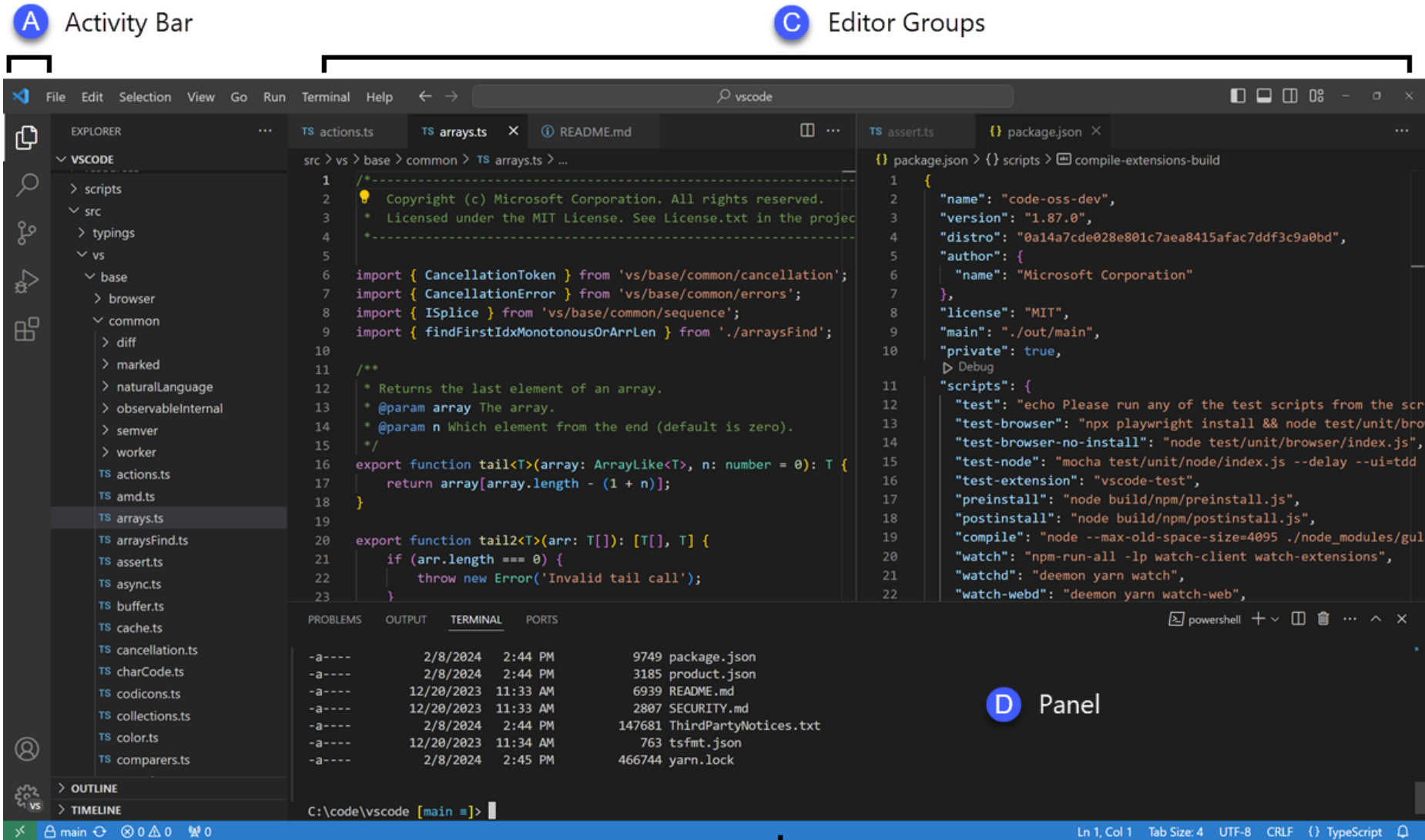
- Introduce VSCode
- Introduce version control and git
- Introduce Markdown for note taking
- Distribute Markdown cheat sheet
- Windows users: install WSL
- Install VSCode

- Professional programmers often use an “Integrated Development Environment” (**IDE**)
 - Write code, browse your files, and even run your program all in the same window
- **Visual Studio Code (VSCode)**: free application from Microsoft
 - Runs on Windows, Mac, Linux—Windows users will run on WSL
 - Supports all popular coding languages
 - Additional plugins for expanded functionality
 - As you write code, VS Code offers suggestions, catches errors, and highlights your code in color to make it easier to read and understand



<https://survey.stackoverflow.co/2025/technology#1-dev-id-es>

VSCode



B Primary Side Bar

E Status Bar

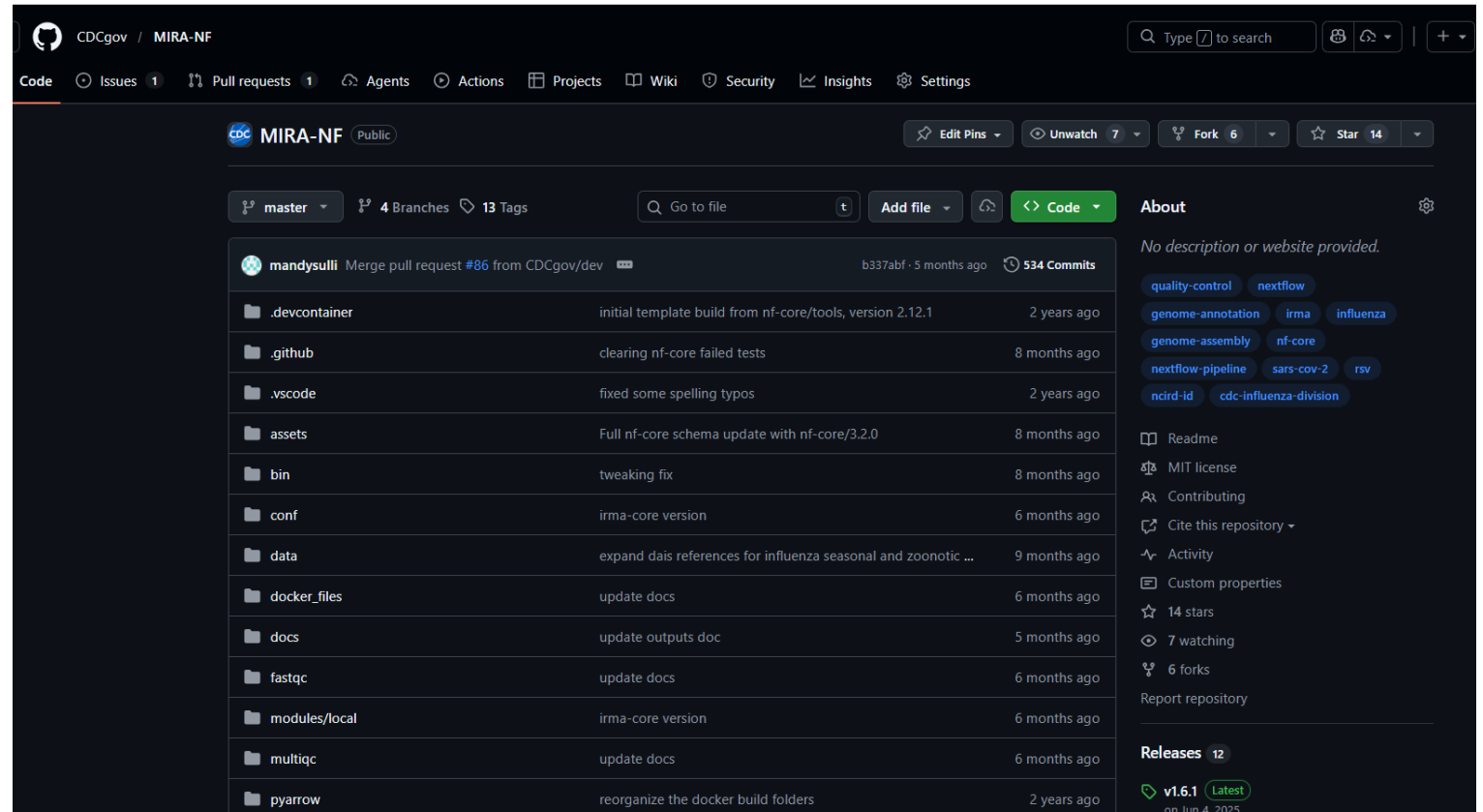
Git

- Git is a version control software running on your local machine
 - Tracks changes to files over time
- Designed for collaboration
 - Multiple people can work on the same codebase
 - Features like issues and milestones allow collaborators to share to-dos and track problems
 - Think of it like a shared document (Microsoft office, Google docs)
- Keeps a history of your work: see what changed, when, and why
- Common in bioinformatics
 - Pipelines, scripts, documentation, and workflows
- Gitlab and Github most common *cloud repositories*



Core Git Concepts

- Repository (repo)
 - A project tracked by Git
- Commit
 - A saved snapshot of changes with a message
- Branch
 - A parallel version of the code for development or testing
- Remote repository
 - Shared copy on platforms like GitHub or GitLab



```
qgx6@rosalind02:MIRA-NF$ git status
On branch dev
Your branch is up to date with 'origin/dev'.

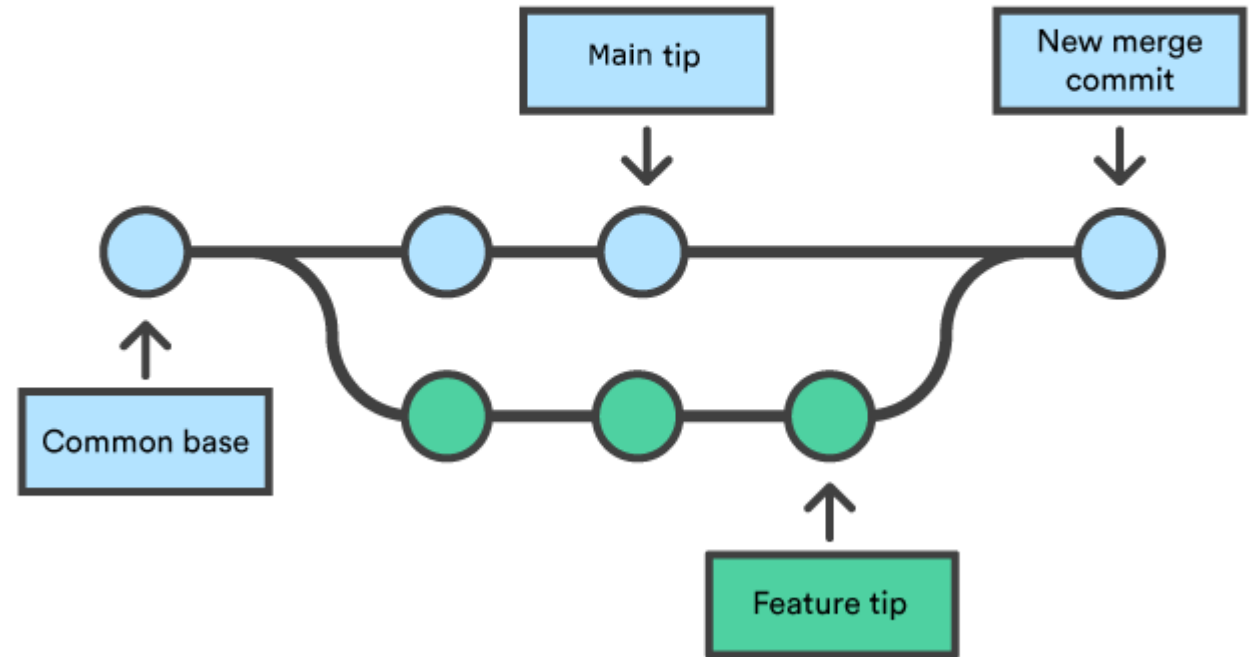
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .MIRA_nextflow.sh

nothing added to commit but untracked files present (use "git add" to track)
```

Git

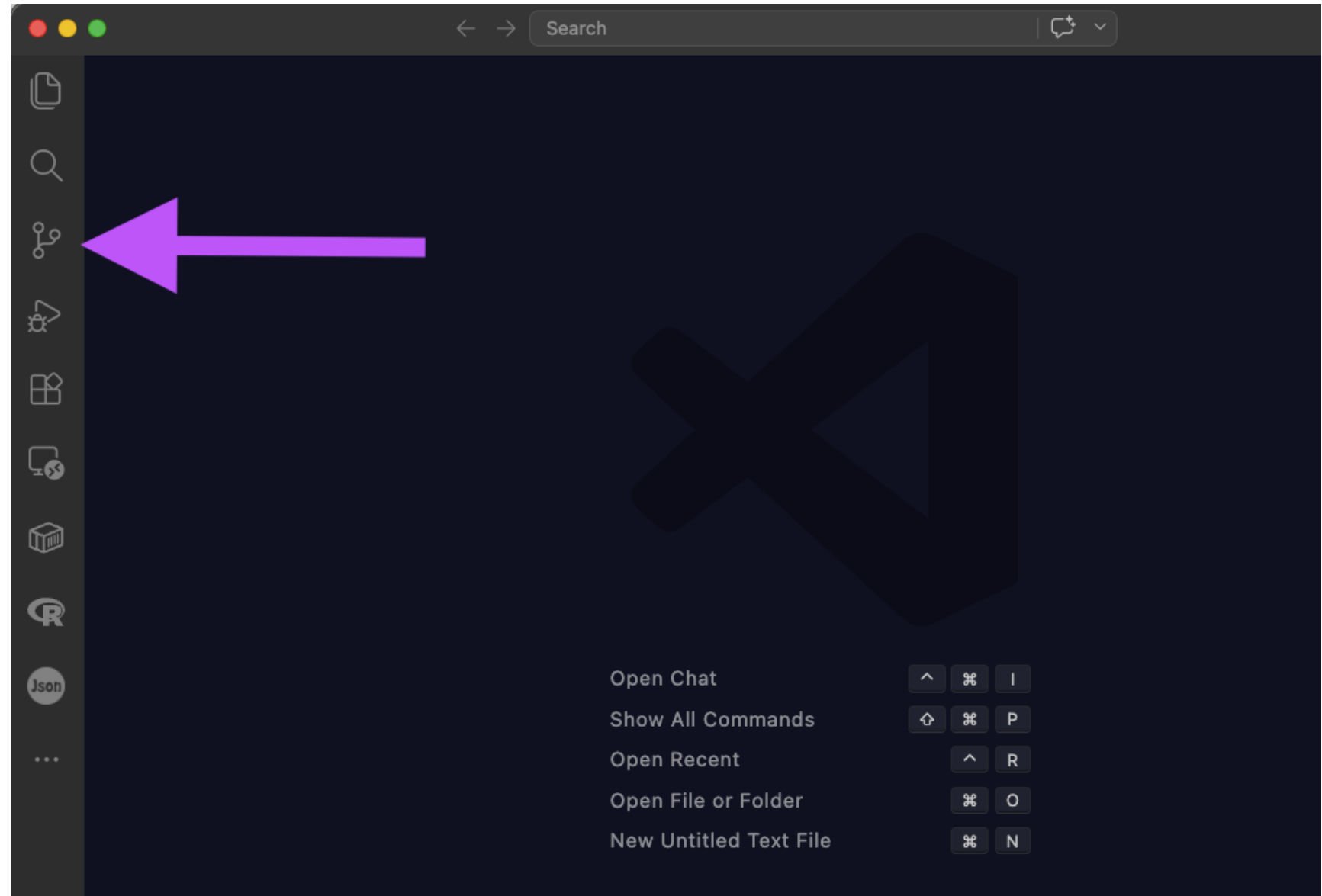
Common Git Commands

- Create or get a repository
 - `git init` — start a repo
 - `git clone <repo>` — copy an existing repo
- Track and save changes
 - `git status` — see changes
 - `git add <file>` — stage changes
 - `git commit -m "message"` — save snapshot
- Sync with others
 - `git pull` — get updates
 - `git push` — share your changes
- Merge or Pull request
 - Combine your changes with the “main” branch



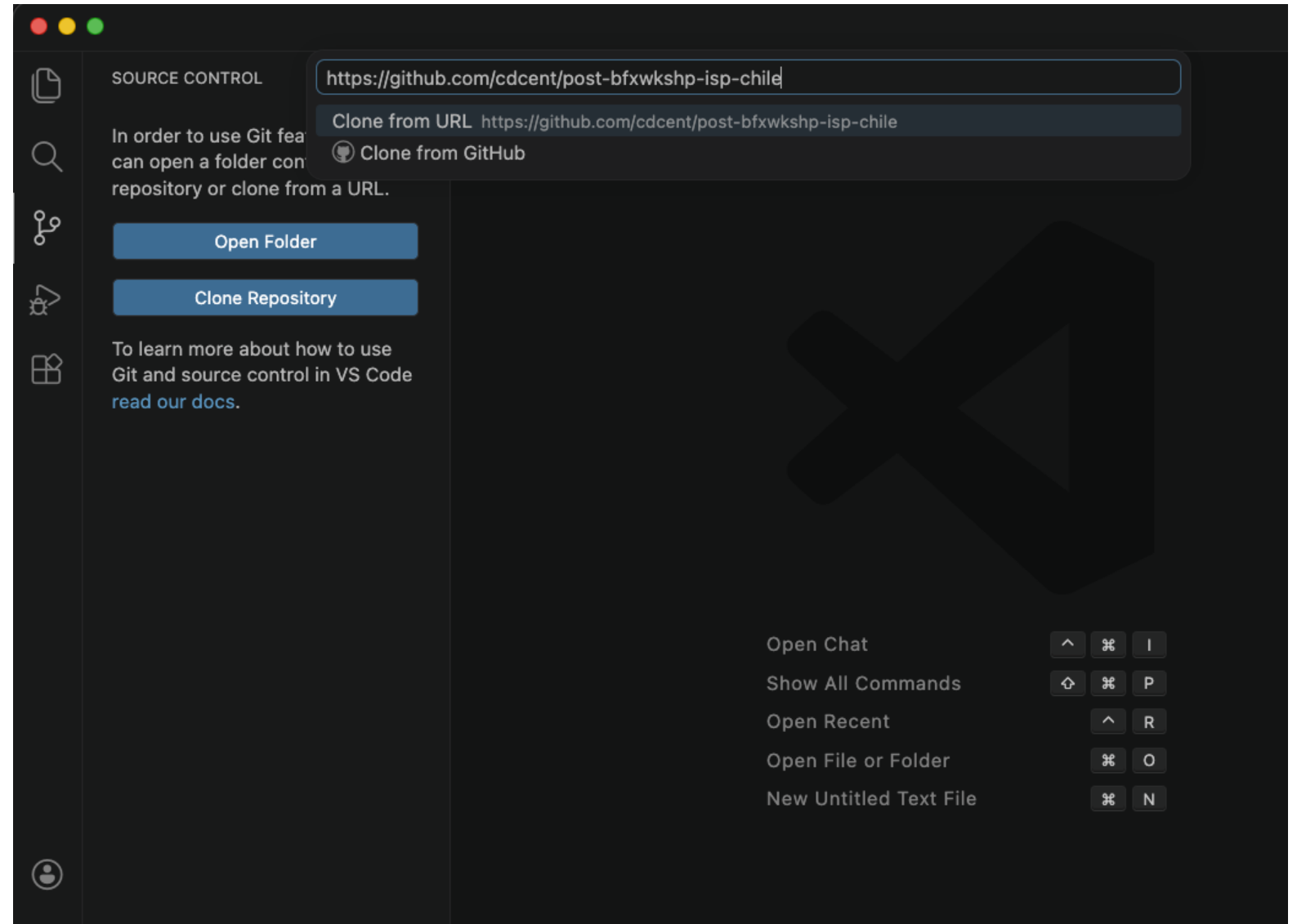
Git in VSCode

Source control tab



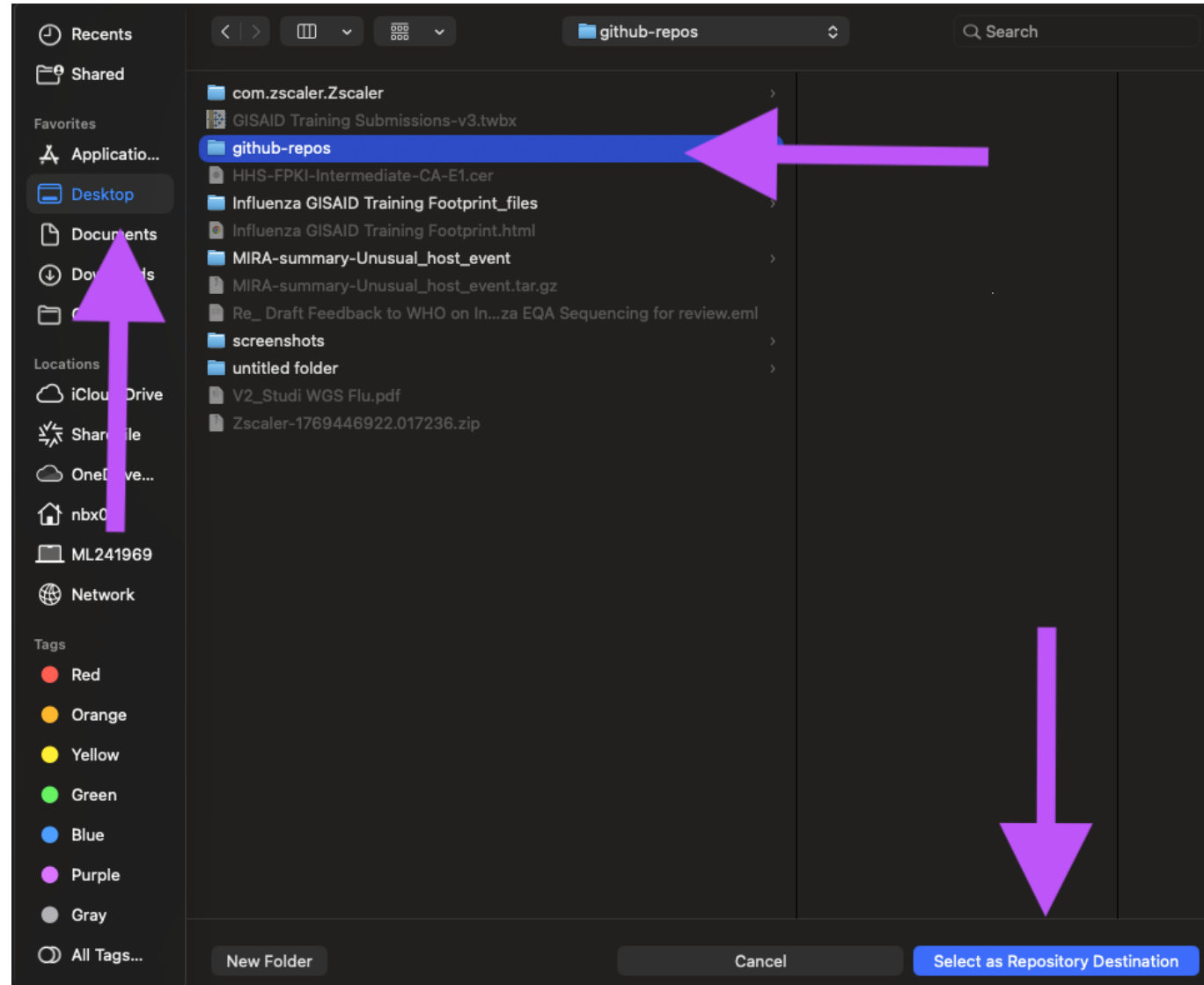
Git in VSCode

Clone repository with URL



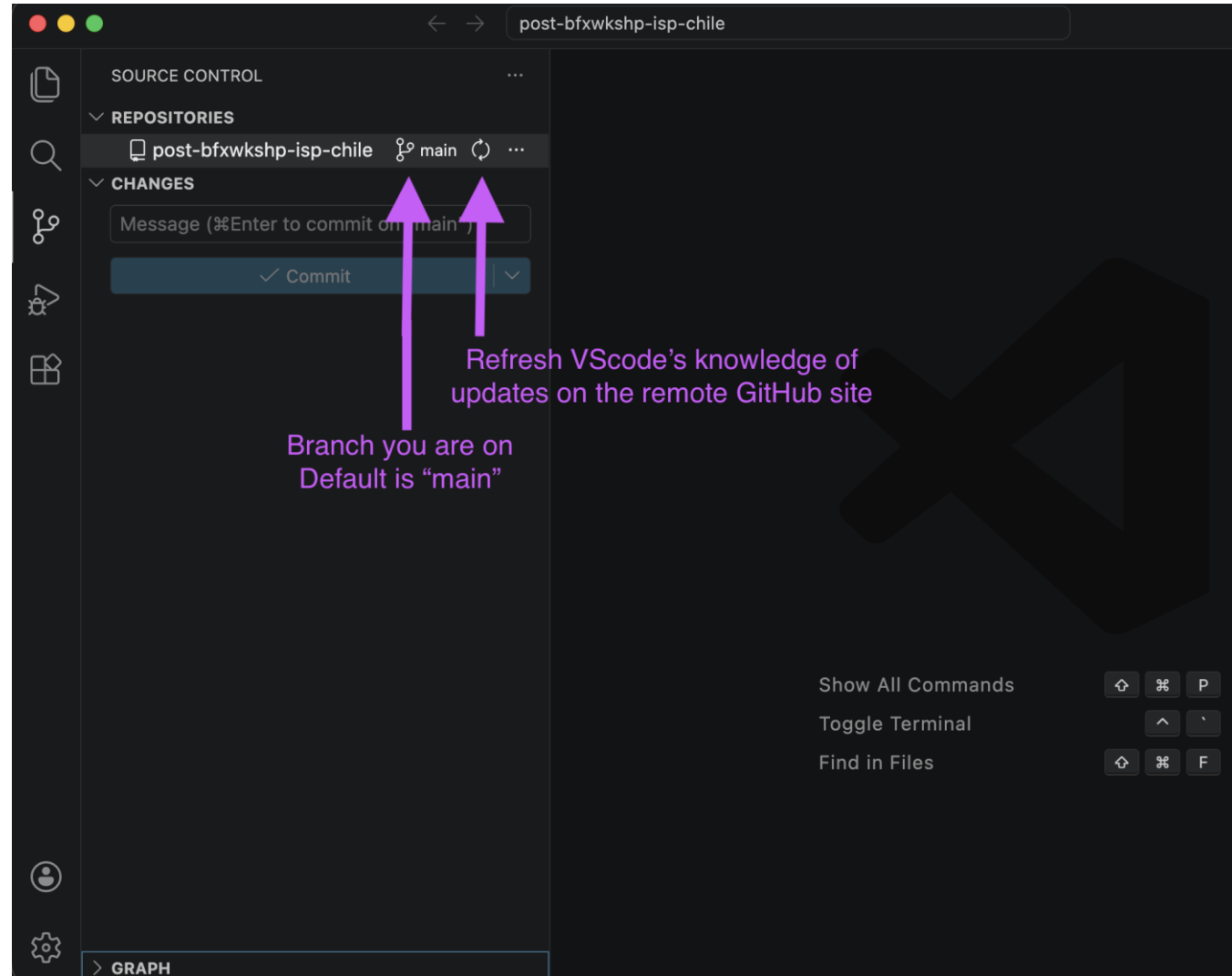
Git in VSCode

Choose a local folder to clone into and click **Select as Repository Destination**. Below, I have created a folder on my Desktop called github-repos to store all my cloned repositories.



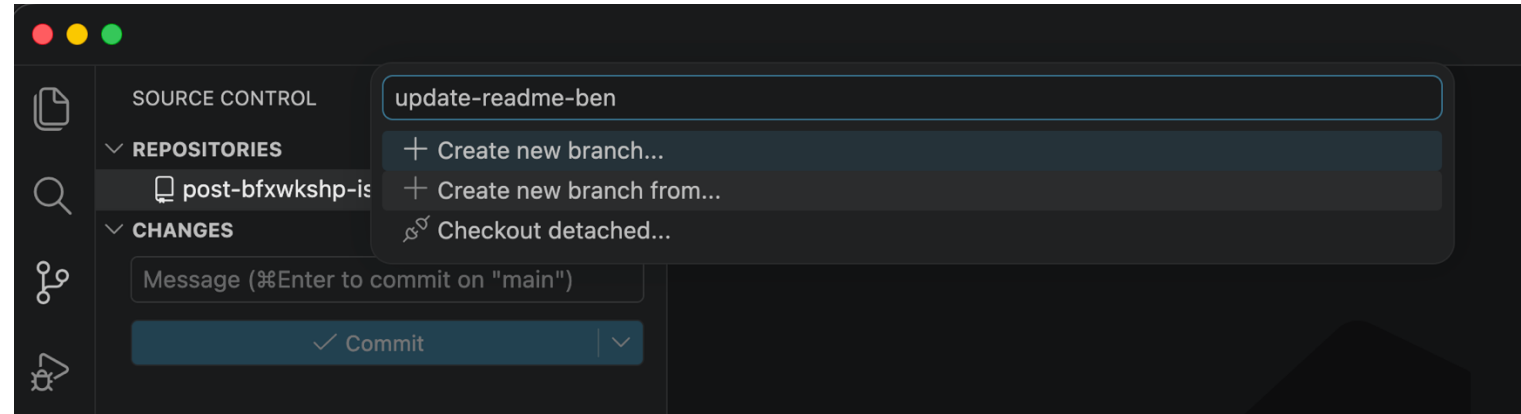
Git in VSCode

In the **Source Control** tab, click the “branch” icon beside “main” at the top of the **REPOSITORIES** panel.

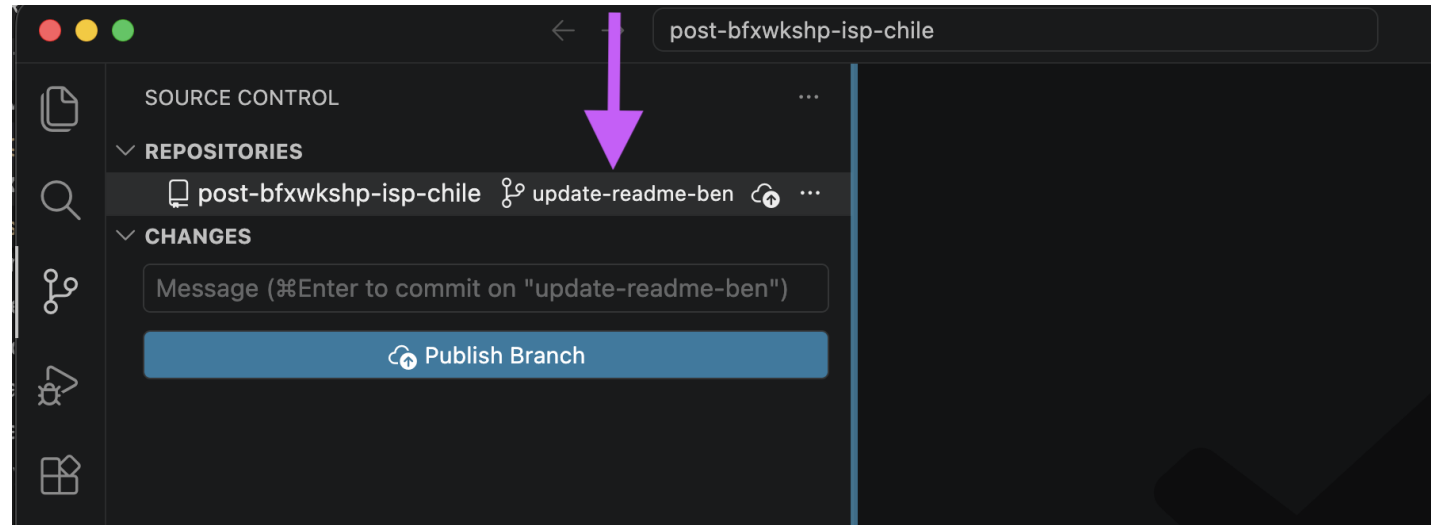


Git in VSCode

Your cursor will move to the input box at the top of the window. Type in a name for your new branch, hit Enter



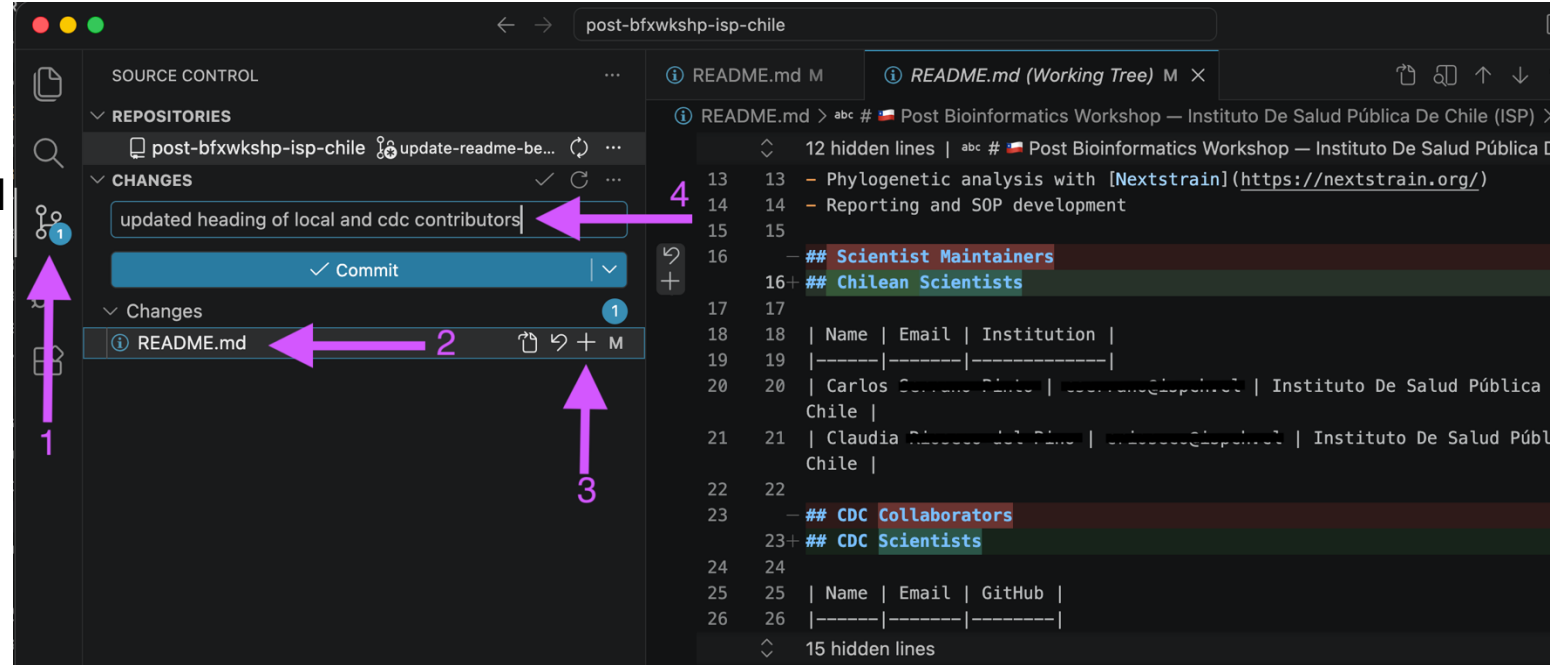
VS Code will create the branch and switch to it automatically. “Publish” the branch to GitHub by clicking the **Publish Branch** button that appears in the status bar at the bottom of the window.



Git in VSCode

After editing and saving a file in the repo via VSCode, Stage Commit, and Push

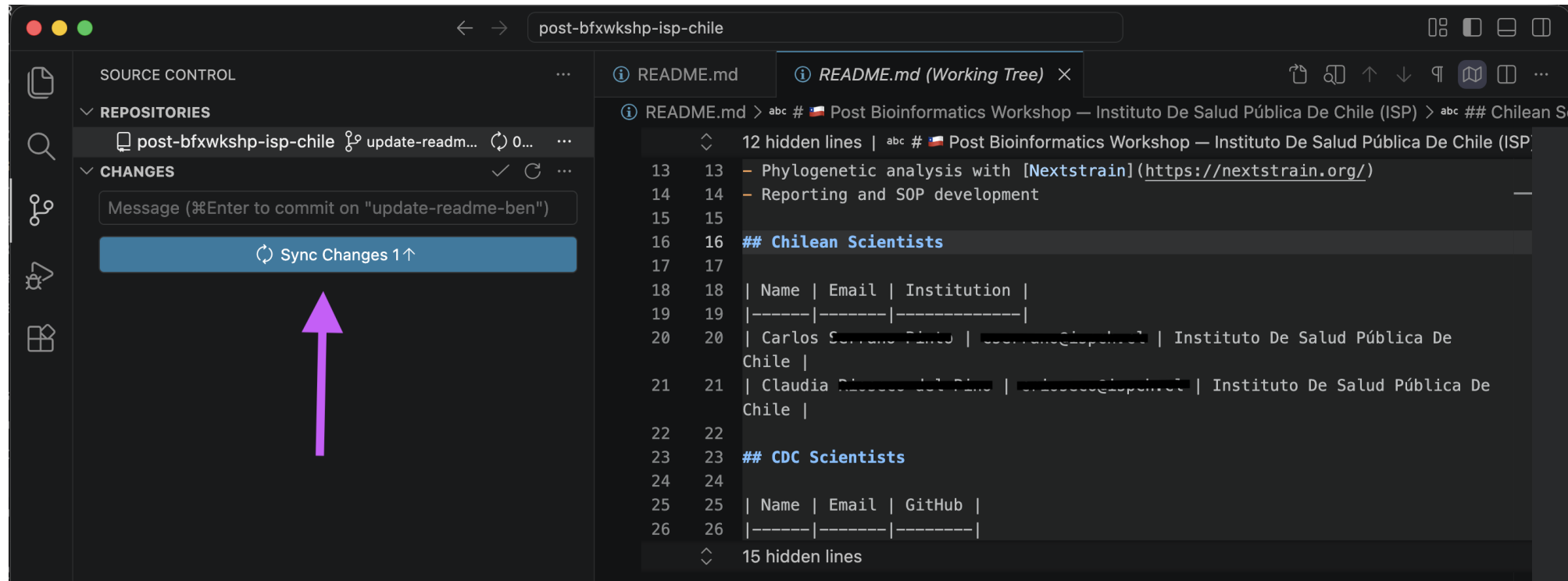
1. Click the **Source Control** tab in the left sidebar.
2. You should see README.md listed under the **Changes** section. Clicking on this file will display a “diff” view showing your changes. Red highlights indicate removed text, and green highlights indicate added text.
3. Hover over README.md and click the + (Stage Changes) icon to move it to the **Staged Changes** section.
4. Type a commit message in the **Message** text box at the top of the Source Control panel, e.g., updated heading of local and cdc contributors.
5. • 5. Click the **Commit** button (checkmark icon) to commit your staged changes.



Git in VSCode

After editing and saving a file in the repo via VSCode, Stage Commit, and Push

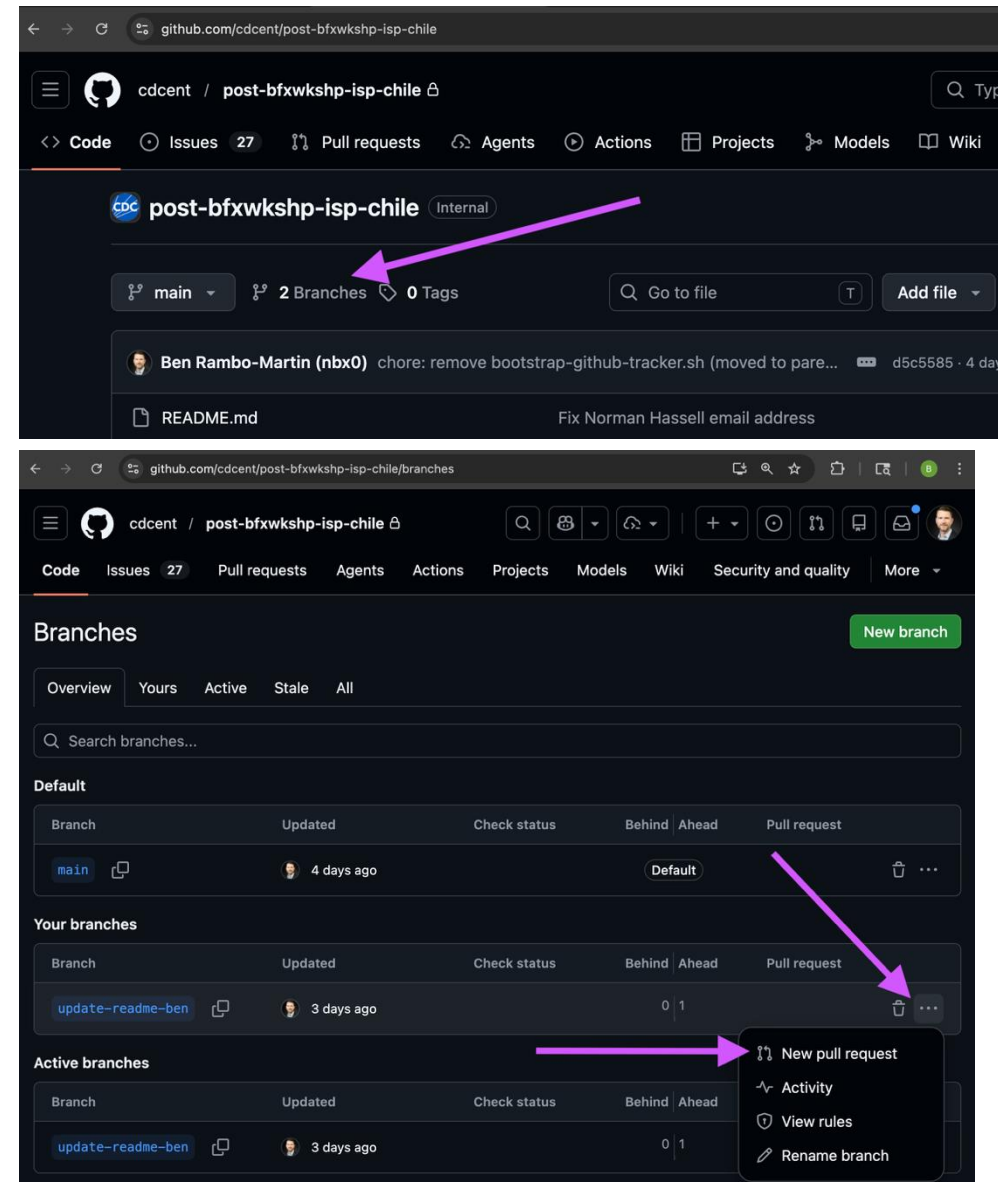
The blue button will now say **Sync Changes**. Click it to push your commit to GitHub. You may be prompted to sign in to GitHub if you haven't already



Git in VSCode

Create a Pull Request and Merge (GitHub Website, not VS Code)

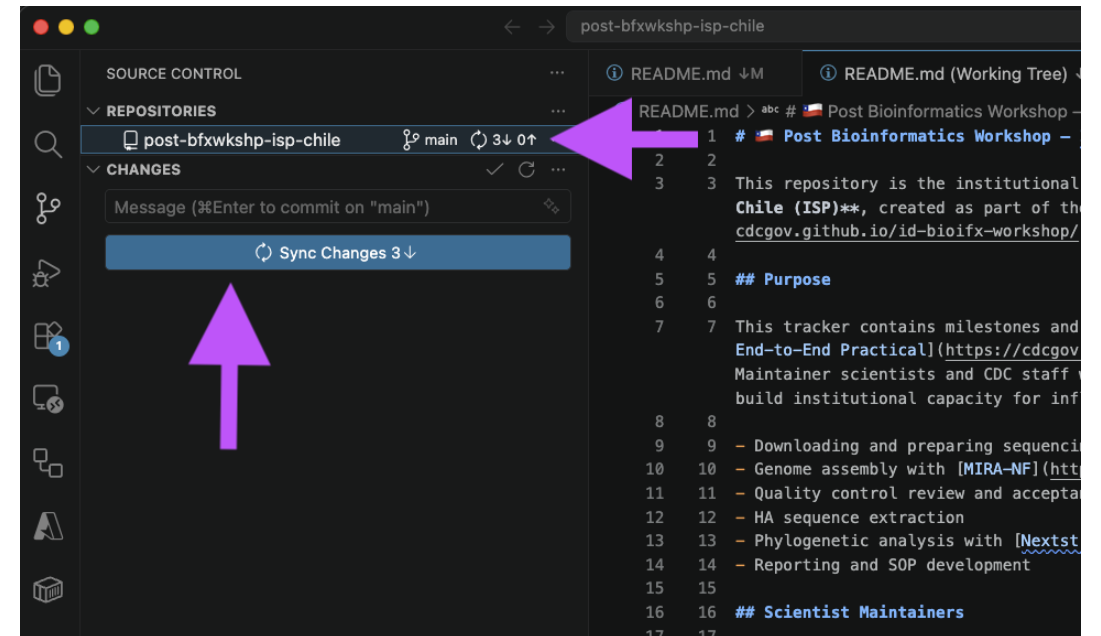
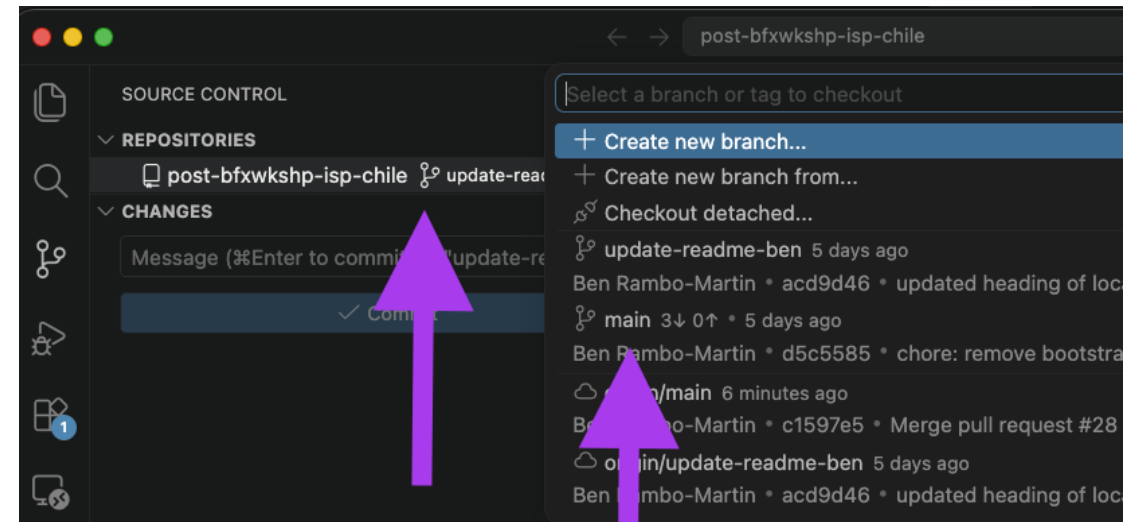
1. Open your web browser and navigate to your repository on github.com.
2. Click on the **Branches** tab, then find your branch in the list and click the three dots next to it and click **New pull request**.
3. Review the changes, add a title and description if needed, then click **Create pull request**
4. For these workshop repos, CDC administrators will complete the merge



Git in VSCode

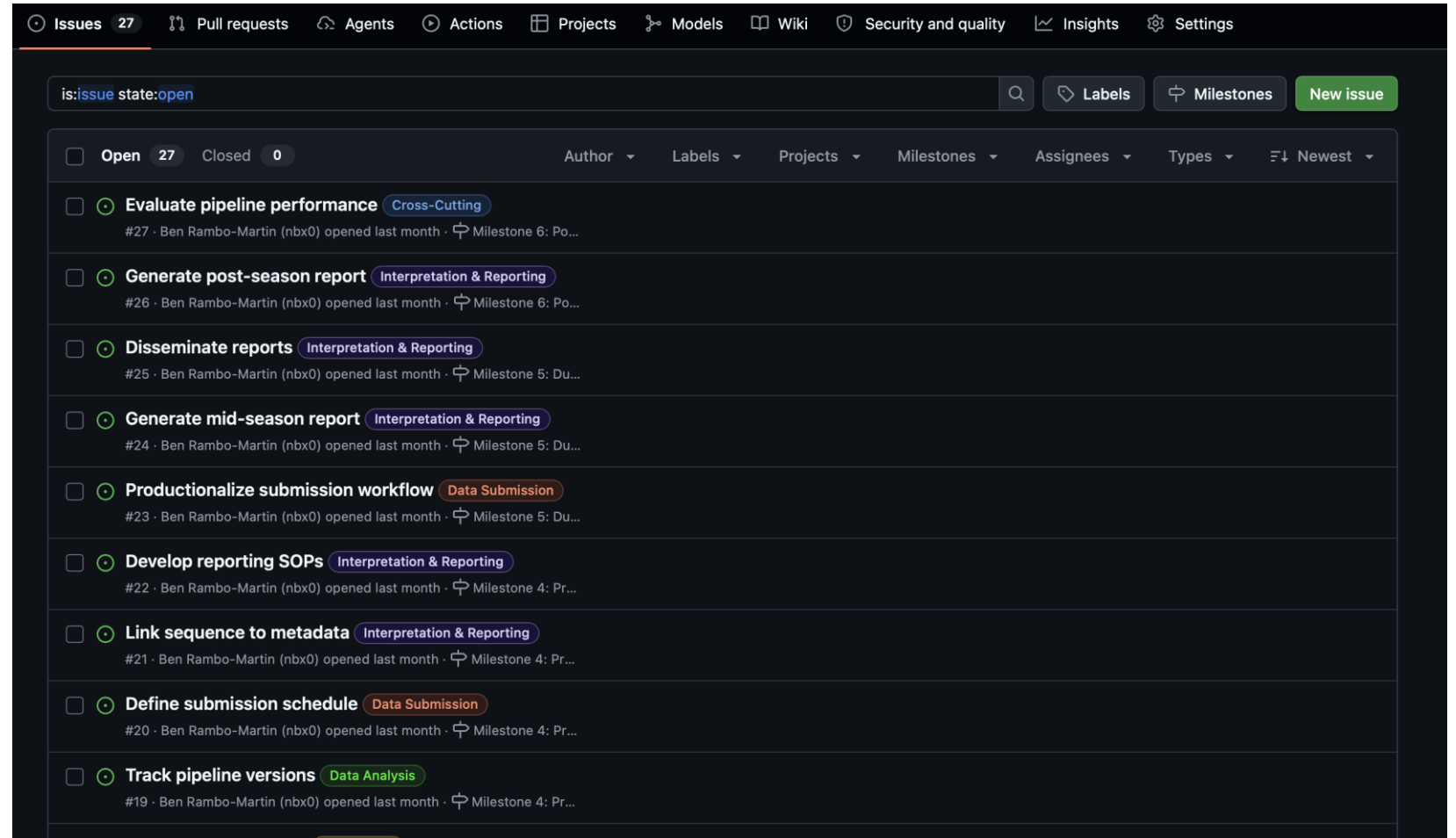
Finally: pull changes locally

1. Back in VS Code, click the **Source Control** tab. Click the “branch” icon at the top of the **REPOSITORIES** panel to switch branches. Select main to switch back to the main branch
2. Click **Sync Changes** to pull the latest changes from GitHub, which will include the merged changes from your branch.



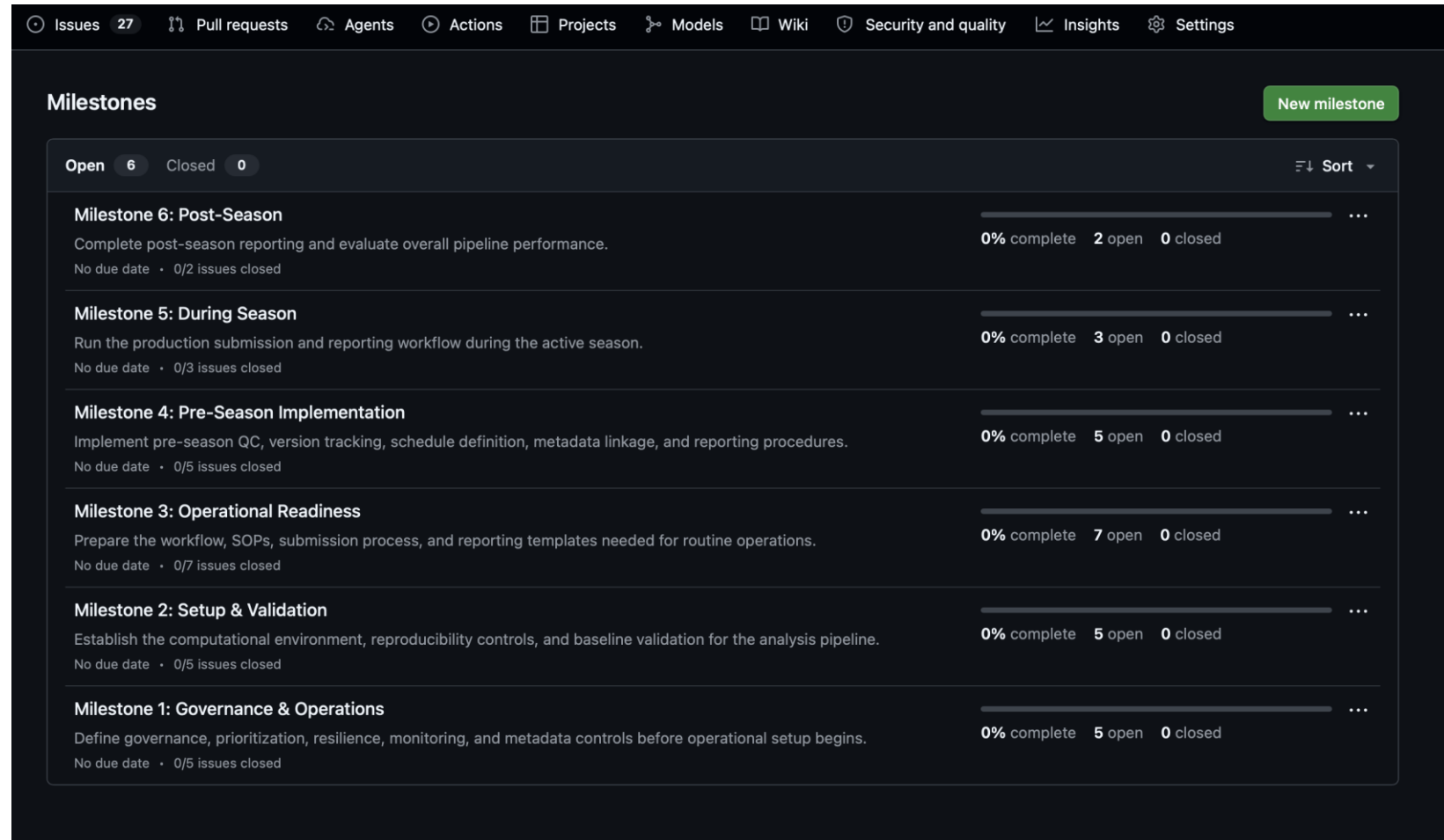
Git issues and milestones

- Git issues (on web) are a to-do list
 - Search
 - Assign
 - Label
 - Leave comments, conversation
 - @ other users
 - “Close” when complete



Git issues and milestones

- Milestones are a collection of issues
 - Bin related issues
 - Track overall progress



The screenshot displays the GitHub 'Milestones' interface. At the top, a navigation bar includes links for Issues (27), Pull requests, Agents, Actions, Projects, Models, Wiki, Security and quality, Insights, and Settings. A 'New milestone' button is located in the top right corner. Below the navigation bar, the 'Milestones' section is titled, with filters for 'Open' (6) and 'Closed' (0) issues, and a 'Sort' dropdown menu. The main content area lists six milestones, each with a title, description, progress bar, and issue counts (0% complete, X open, Y closed). The milestones are ordered from 6 at the top to 1 at the bottom.

Milestone	Description	Progress	Open	Closed
Milestone 6: Post-Season	Complete post-season reporting and evaluate overall pipeline performance.	0% complete	2	0
Milestone 5: During Season	Run the production submission and reporting workflow during the active season.	0% complete	3	0
Milestone 4: Pre-Season Implementation	Implement pre-season QC, version tracking, schedule definition, metadata linkage, and reporting procedures.	0% complete	5	0
Milestone 3: Operational Readiness	Prepare the workflow, SOPs, submission process, and reporting templates needed for routine operations.	0% complete	7	0
Milestone 2: Setup & Validation	Establish the computational environment, reproducibility controls, and baseline validation for the analysis pipeline.	0% complete	5	0
Milestone 1: Governance & Operations	Define governance, prioritization, resilience, monitoring, and metadata controls before operational setup begins.	0% complete	5	0

Markdown

- A simple way to format text using plain, readable symbols like asterisks and hashtags
- No special software needed — it can be written in any basic text editor
- Very easy to learn, with a minimal set of rules
- Used widely across the web, including GitHub, blogs, and note-taking apps
- Automatically converts into clean, formatted web pages or documents when **rendered**
- Markdown files end with the extension .md

Markdown

- # Heading 1, ## Heading 2, ### Heading 3
 - create headings with pound signs
 - 1 is biggest
- **bold text**
 - wrap in double asterisks for **bold**
- *italicized text*
 - wrap in single asterisks for *italic*
- > blockquote
 - add a > to create an indented quote block
- - bullet point list

Markdown

```
1 # Heading 1
2 ## Heading 2
3 ### Heading 3
4
5 **bold text**
6
7 italic text
8
9 >block quote
10
11 - bullet
12 - point
13 - list
```

Heading 1

Heading 2

Heading 3

bold text

italic text

block quote

- bullet
- point
- list

Markdown

- `1. 2. 3.` : start lines with numbers for an ordered list
- ``inline code`` : wrap in backticks for inline code
- ````` on its own line : opens and closes a fenced code block
- `[link text](url)` : Create a clickable link
- `![alt text](image.jpg)` : embed an image (same as a link, but with !)
- `---` : three dashed lines make a horizontal dividing line

Markdown

markdown-vscode.md

```
1 1. numeric
2 2. list
3 3. now
4
5 `one line of code`
6
7 ```
8 multiple
9 lines
10 of
11 code
12 ```
13
14 [Go to google](www.google.com)
15
16 ![CDC](CDClogo.png)
17
18 ---
```

1. numeric
2. list
3. now

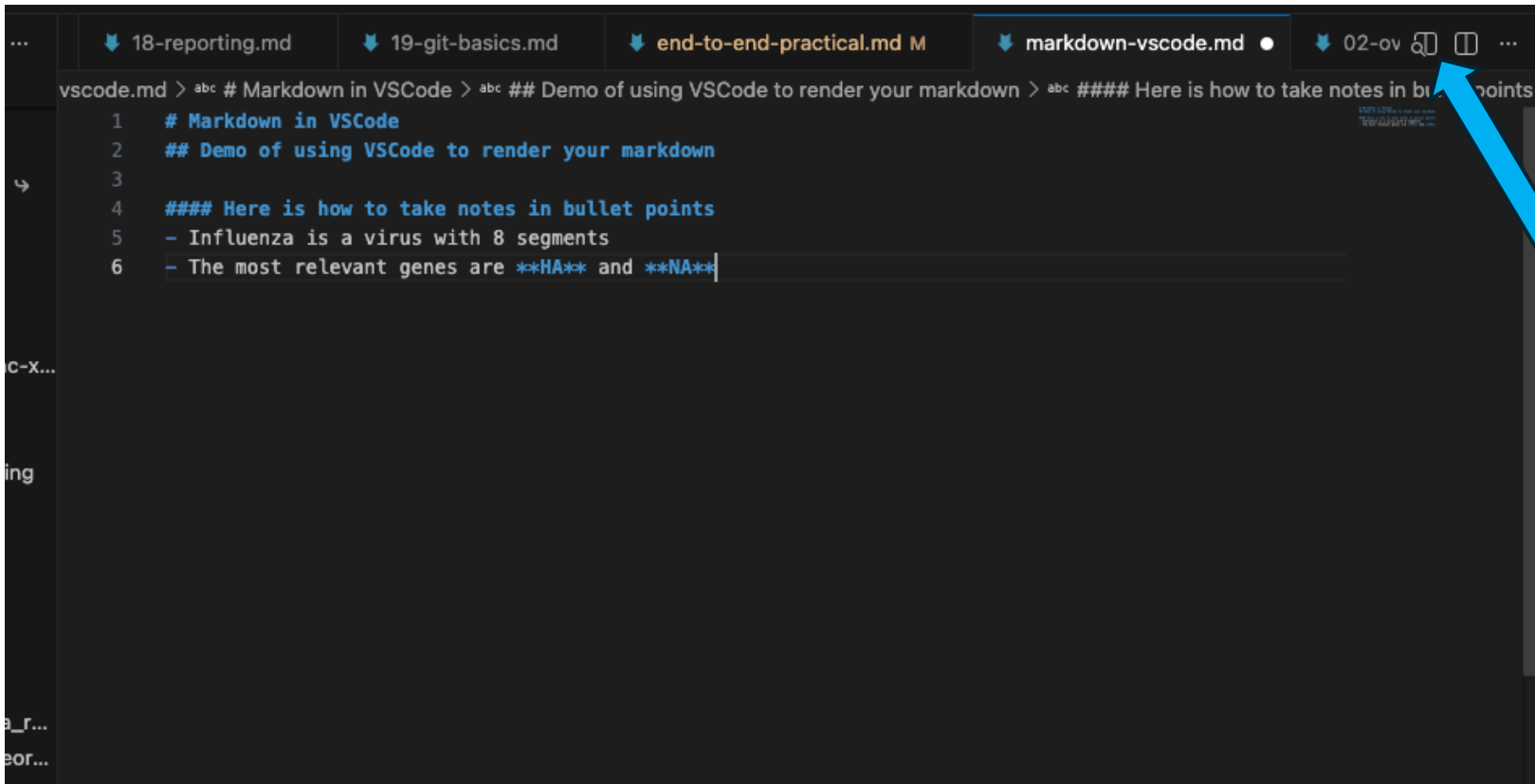
one line of code

```
multiple
lines
of
code
```

[Go to google](http://www.google.com)



Markdown in VSCode



The screenshot shows the VS Code editor interface with a dark theme. The top bar displays several open files: '18-reporting.md', '19-git-basics.md', 'end-to-end-practical.md M', 'markdown-vscode.md', and '02-ov'. The 'markdown-vscode.md' file is the active editor. The editor content shows a markdown document with the following text:

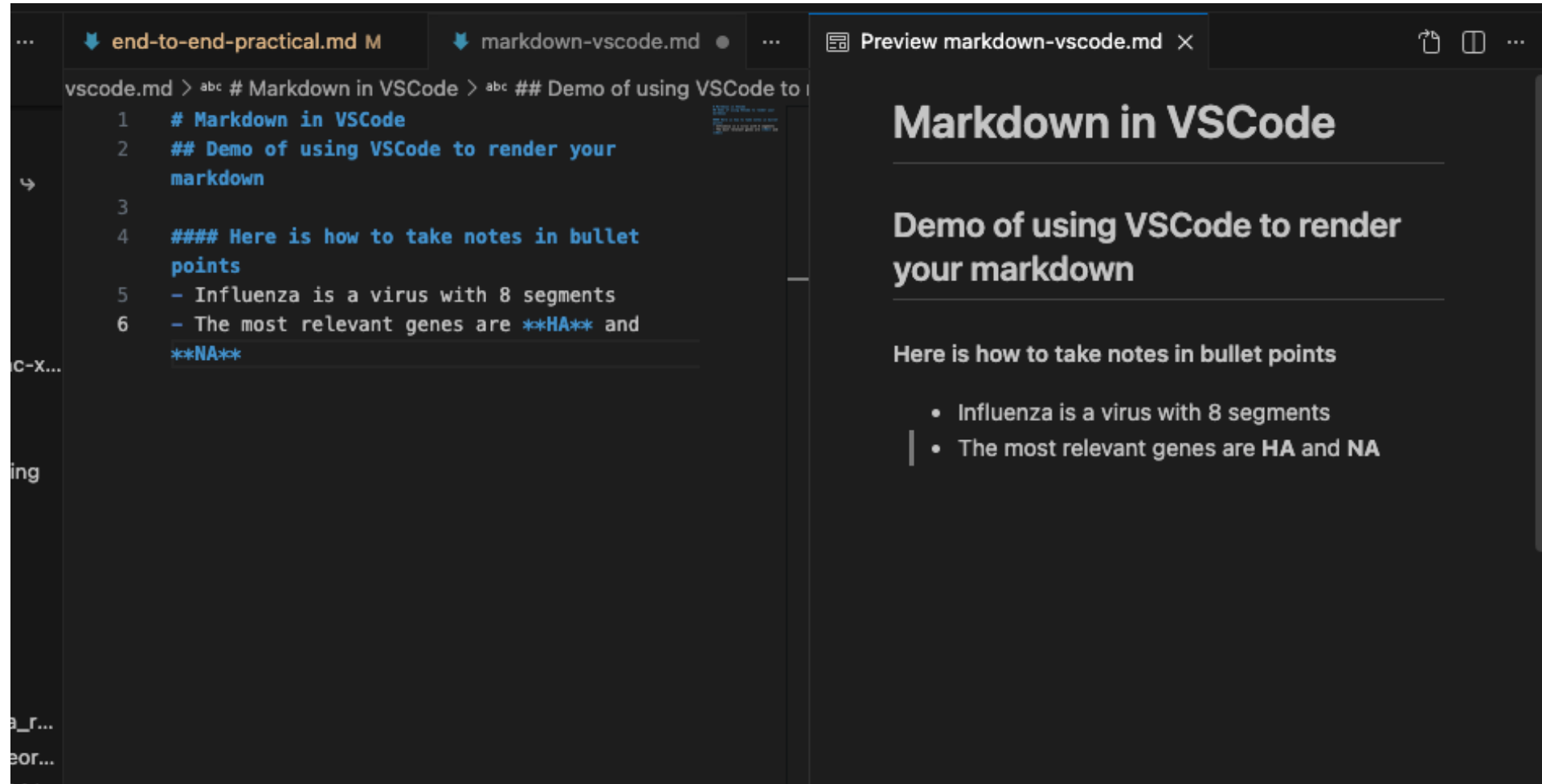
```
vscode.md > abc # Markdown in VSCode > abc ## Demo of using VSCode to render your markdown > abc #### Here is how to take notes in bullet points
1 # Markdown in VSCode
2 ## Demo of using VSCode to render your markdown
3
4 #### Here is how to take notes in bullet points
5 - Influenza is a virus with 8 segments
6 - The most relevant genes are **HA** and **NA**
```

A blue arrow points from the text 'Click the magnifying glass to render a preview' to the magnifying glass icon in the top right corner of the editor window.

Open a .md file

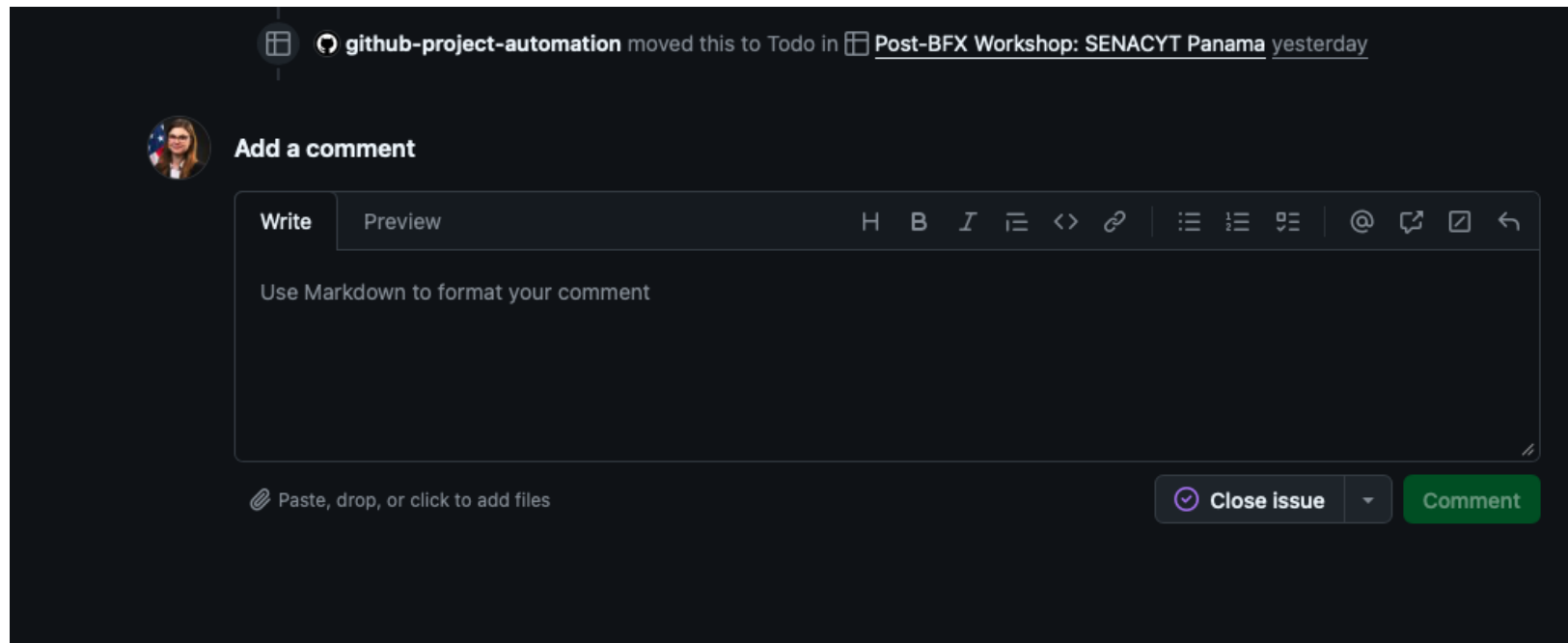
Click the magnifying glass to render a preview

Markdown in VSCode



Git renders Markdown

- Readme files
- Git issues
- Comments on gitlab
- Documentation
- Even our website for this course was written in Markdown!



Markdown in VSCode: note-taking

- This week, take notes in VSCode with Markdown
- It is worthwhile for future reporting
- Cleaner way to write notes about code and commands
- You can commit notes to git where they can be backed up daily
- Collaborate/combine notes with others from your lab
- In the practical, open a new file called Day1_notes.md



Practical: install VSCode

1. If you are on a Windows machine without WSL (Ubuntu), have a trainer help you through the install
2. Go to code.visualstudio.com and install VSCode
3. Clone your workshop repo
4. Follow instructions on course website to branch, start a note-taking markdown file
5. Merge !